Math Inquirers Level 2: skill additions (tutor sheet)

Extra items for early finishers and for tutors to set live. They are additions: nothing here is in the printed workbook. **Slide** items are on an optional slide in the lesson deck (after the Go further walk-through). **Bank** items are not on a slide: read or write them for the pair. Children write the answer on their own paper and say it together on three. Nothing is held up to the camera. Never announce who is working on which item.

Lesson 2: Missing Digit Mysteries

SLIDE Tool: Try something simpler **Pairing sums: first and last make the same total** (L2-PAIR1)

Add these numbers. Look for pairs that make the same total.

a) 4 + 9 + 14 + 19 + 24 + 29 + 34 + 39 = [ ]

b) Add all the whole numbers from 11 to 20.

Answer. a) 4 + 39 = 43, 9 + 34 = 43, 14 + 29 = 43, 19 + 24 = 43: four pairs of 43 = 172. b) 11 + 20 = 31, and there are 5 pairs: 155.

Say. Write the list forwards, then write it again backwards underneath. Add the two numbers that sit on top of each other.

Ask. What do you notice about the first and the last? And the second and the second last? How many pairs are there?

Listen for. "Every pair makes the same total." Check that they count the pairs, not the numbers.

BANK Tool: Try something simpler **Pairing sums: save every day** (L2-PAIR2)

Add the numbers.

a) 40 + 80 + 120 + 160 + 200 + 240 = [ ]

b) Mei saves $5 on day 1, $10 on day 2, $15 on day 3, and so on, up to $50 on day 10. How much has she saved after 10 days?

Answer. a) 40 + 240 = 280 and there are 3 pairs: 840. b) 5 + 50 = 55 and there are 5 pairs: $275.

Say. Try something simpler: do 4 numbers first.

Ask. How many pairs? Does it still work if there is an odd number of numbers?

Listen for. Pairs times total.

SLIDE Tool: Guess and check **Letter-digit sums** (L2-LETTER1)

Each letter stands for one digit. Different letters stand for different digits. Find A, B and C.

A = [ ] B = [ ] C = [ ]

Answer. A = 1, B = 9, C = 7 (18 + 79 = 97). Ones: 8 + 9 = 17, so C = 7 and carry 1. Tens: A + 7 + 1 = B, so A + 8 = B. A digit stops at 9, so only A = 1 and B = 9 fit.

Say. Start with the ones column, just like the missing-digit sums.

Ask. What must 8 + 9 make in the ones? What is the carry? Which letter is it?

Listen for. The ones column shows C = 7 and a carry. Then test the tens. Ask them to check the whole sum.

Lesson 4: Pattern Machines

SLIDE Tool: Make a table **Sequences: the gap grows by 1 each time** (L2-SEQ1)

Ali writes a pattern: 6, 7, 9, 12, 16, __, __

a) Write the next two numbers.

b) Will 40 be in the pattern?

Answer. a) 21 and 27 (the gaps are 1, 2, 3, 4, 5, 6). b) No: the pattern goes 27, 34, 42 and jumps over 40.

Say. This pattern does not add the same number. Make a table and put the gaps in a row of their own.

Ask. What are the gaps? How does the gap change? What is the next gap?

Listen for. "The gap goes up by 1."

SLIDE Tool: Spot the pattern **Sequences: add the two numbers before** (L2-SEQ2)

Mei writes a pattern: 2, 4, 6, 10, 16, __, __

a) Write the next two numbers.

b) What is the 9th number?

Answer. a) 26 and 42 (6 + 10 = 16, 10 + 16 = 26, 16 + 26 = 42). b) 110: the numbers are 2, 4, 6, 10, 16, 26, 42, 68, 110.

Say. Try the gap row. Does it look like the numbers?

Ask. What is 2 + 4? What is 4 + 6? What would the next number be?

Listen for. "Add the two numbers before." Ask them to write the whole list to the 9th number.

BANK Tool: Spot the pattern **Sequences: two rules take turns** (L2-SEQ3)

Start at 20. Add 10, then take away 4, then add 10, then take away 4, and keep going.

a) Write the first 7 numbers.

b) What is the 9th number?

Answer. a) 20, 30, 26, 36, 32, 42, 38. b) 44.

Say. Treat each pair of steps as one jump of 6.

Ask. What do the numbers in odd places do? In even places?

Listen for. "Every two steps it goes up 6."

BANK Tool: Make a table **Sequences: the gap gets smaller** (L2-SEQ4)

Siti counts down: 100, 90, 81, 73, 66, __, __

a) Write the next two numbers.

b) What is the gap each time?

Answer. a) 60 and 55. b) The gaps are 10, 9, 8, 7, 6, 5: each gap is 1 less than the one before.

Say. Draw a table with a row for the gap.

Ask. What is the next gap? How do you know?

Listen for. "The gaps go down 1 each time."

Lesson 5: Mystery Number Machines

SLIDE Tool: Draw it **Ages: the gap never changes** (L2-AGE1)

Read each story. Draw a bar for each person.

a) Ali is 4 years older than his sister. Their ages add up to 20. How old is each?

b) Mum is 28 years older than Mei. Mei is 7 now. How old will each be in 5 years?

Answer. a) Take away the 4 extra years: 16 left, shared equally is 8. Sister 8, Ali 12 (8 + 12 = 20). b) Mei will be 12; Mum is 35 now, so 40 in 5 years. The gap is still 28.

Say. For part a, take away the extra years first, then share what is left. The same take-away-the-extra idea as the worksheet.

Ask. Which person is older? By how many years? What do you do with the extra 4?

Listen for. "Both get older at the same speed, so the gap stays." For a), a Guess and check that tests both clues is also fine.

BANK Tool: Make a table **Ages: when will it happen?** (L2-AGE2)

Ravi is 9 years old. Mei is 5 years old.

a) Ravi will be 12 years old in ___ years.

b) Mei will be ___ years old then.

c) Ravi was ___ years old when Mei was born.

Answer. a) 3 years. b) 8 years old. c) 4 years old.

Say. Make a small table: year, Ravi, Mei.

Ask. What number stays the same in every row?

Listen for. The gap of 4.

Lesson 8: Shape Builders

SLIDE Tool: Act it out **Cubes in the corner of a room** (L2-CORNER1)

Cubes are stacked in the corner of a room, touching both walls. The bottom layer is 3 rows of 3. The next layer is 2 rows of 2. The top layer is 1 cube.

a) The bottom layer has ___ cubes.

b) The whole stack has ___ cubes.

c) ___ cubes touch the left wall.

Answer. a) 9. b) 14 (9 + 4 + 1). c) 6 (3 in the bottom layer, 2 in the next and 1 on top).

Say. Build it layer by layer with cubes, then look at one wall.

Ask. Which cubes touch the wall in the bottom layer? In the next layer?

Listen for. "3, then 2, then 1."

BANK Tool: Make a table **Cubes in a corner: bigger** (L2-CORNER2)

A cube corner has layers of 4 by 4, 3 by 3, 2 by 2 and 1 cube.

a) How many cubes are there in all?

b) How many cubes touch the floor?

Answer. a) 30 (16 + 9 + 4 + 1). b) 16.

Say. Make a table: layer, size, cubes.

Ask. What pattern do the layer numbers make?

Listen for. Square numbers: 16, 9, 4, 1.

Lesson 9: Measure Masters

SLIDE Tool: Guess and check **Picture equations: shapes stand for numbers** (L2-PIC1)

Each shape stands for a number. The same shape is the same number.

a) triangle = [ ] b) circle = [ ] c) square = [ ]

d) triangle + circle + square = [ ]

Answer. a) 9 (27 split into 3 equal parts). b) 6 (21 - 9 = 12, shared by 2). c) 7 (13 - 6). d) 22.

Say. Find the line that tells you the most. Which shape can you find first?

Ask. Which line did you use first and why? How can you check?

Listen for. Three triangles = 27 is the start. Then use it in line 2.

BANK Tool: Make a table **Picture equations: exchange chain** (L2-PIC2)

2 apples weigh the same as 1 melon. 3 melons weigh the same as 1 pumpkin.

a) How many apples weigh the same as 1 pumpkin?

b) 1 pumpkin and 2 melons weigh the same as how many apples?

Answer. a) 6 apples (3 melons, each 2 apples). b) 10 apples (6 + 4).

Say. Draw it as a chain: apple, melon, pumpkin.

Ask. How many apples for one melon? For three melons?

Listen for. "Multiply along the chain."

SLIDE Tool: Spot the pattern **Calendar: weeks and days** (L2-CAL1)

Today is Thursday.

a) What day will it be in 3 weeks?

b) What day will it be in 25 days?

c) What day will it be in 30 days?

Answer. a) Thursday (3 weeks is 21 days, 3 full cycles). b) Monday (25 = 3 weeks and 4 days; Thursday + 4 days is Monday). c) Saturday (30 = 4 weeks and 2 days).

Say. The week is a repeating pattern of 7. Take out whole weeks first.

Ask. How many whole weeks fit into 25 days? How many days are left over?

Listen for. "Remainder" language: 25 days is 3 weeks and 4 days.

BANK Tool: Make a table **Calendar: dates in a month** (L2-CAL2)

A month has 30 days. The 1st of the month is a Tuesday.

a) Which dates are Tuesdays?

b) What day is the 30th?

c) What day is the 1st of next month?

Answer. a) 1, 8, 15, 22 and 29. b) Wednesday. c) Thursday.

Say. Write the dates in rows of 7 under the days of the week.

Ask. What do you notice about the Tuesday dates?

Listen for. They go up by 7.

Lesson 10: Puzzle Day

SLIDE Tool: Draw it **Posts and gaps** (L2-POST1)

A straight path is 20 m long. A lamp post stands at the start, at the end, and every 5 m in between.

a) How many lamp posts are there?

b) A fence has 7 posts in a straight line, 3 m apart. How long is the fence?

Answer. a) 5 posts (4 gaps of 5 m). b) 18 m (6 gaps of 3 m).

Say. Draw a dot for each post and a line for each gap.

Ask. Are there more posts or more gaps? Why?

Listen for. "One more post than gaps" in a line.

SLIDE Tool: Draw it **Places in a queue** (L2-QUEUE1)

Counting from the front of a queue, Ravi is number 5. Counting from the back, he is number 4.

Find the number of children in the queue.

Answer. 8 children. Ravi is counted twice (5 + 4 = 9), so take 1 away.

Say. Draw the queue and mark Ravi.

Ask. Who is in front of Ravi? Behind him? Did you count Ravi twice?

Listen for. "Ravi is counted in both numbers."

BANK Tool: Draw it **Posts and gaps: round a pond** (L2-POST2)

8 lamps stand round a pond, the same distance apart. It is 4 m from one lamp to the next.

a) How many gaps are there?

b) How long is the path all the way round?

Answer. a) 8 gaps. b) 32 m.

Say. In a ring there is no end post.

Ask. Is it the same as a straight path? Why not?

Listen for. Gaps equal lamps in a ring.

SLIDE Tool: Act it out **Worst case: how many to be sure?** (L2-WORST1)

A bag has 6 red counters and 4 blue counters. You take counters out without looking.

a) How many must you take to be sure you have 2 red counters?

b) How many must you take to be sure you have one counter of each colour?

Answer. a) 6 (the worst case is 4 blue first, then 2 red). b) 7 (the worst case is all 6 red first, then 1 blue).

Say. Be unlucky on purpose. What is the worst that can happen?

Ask. What is the unluckiest draw? What counter must come next?

Listen for. "The unlucky draw is all of the other colour first."

BANK Tool: Act it out **Worst case: socks** (L2-WORST2)

A drawer has 4 black socks, 4 white socks and 4 grey socks. You take socks out without looking.

a) How many socks must you take to be sure you have a matching pair?

b) How many socks must you take to be sure you have 2 black socks?

Answer. a) 4 (one of each colour is 3 socks; the 4th must match). b) 10 (4 white and 4 grey first, then 2 black).

Say. Act it out with coloured counters.

Ask. What is the most socks you can take with no pair?

Listen for. "One of each colour, then one more."